

Lecture 1: What is Data Science?

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TL;DR

Read literally, *data science* means science done with data, which covers almost every empirical field. This chapter looks instead at who claims the title and who does not, and finds the difference in whether a person works on the methods or only uses them. That gives the definition used here: the field that develops the apparatus for doing science with data. The chapter follows that apparatus down two lineages, sets out what the definition requires of a data scientist, and argues that a skill-based account of the occupation is what generative AI puts at risk.

AI use declaration. We used Claude Opus 5 to research as well as polish the writing in the notes.
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1 Motivation

Consider three researchers: (i) a landscape ecologist records migratory birds at different wetlands to study what sustains their biodiversity, (ii) an astronomer studies starquakes using telescope readings, and (iii) an epidemiologist links hospital records to identify the risk of readmission. All three collect and analyze data, and all three write code or could write it. None of them is obviously a data scientist.

The answer matters for building a curriculum, and for saying what a data scientist adds to a project that already has a physician and a software engineer on it. Data science has many courses and job titles, and less agreement about what students are meant to learn from them.

2 Science done with Data?

Data and *science* are both ordinary nouns, and together they mean science done using data. All three researchers qualify under that reading, since the ecologist works from field counts, the astronomer from telescope readings, and the epidemiologist from discharge records.

So does nearly everyone else. A demographer, a linguist, and a materials scientist all reason from recorded observations. Empirical science is science done with data, so this reading is close to a tautology, and every empirical field falls inside it.

Who uses the title is a better guide than what the words mean. An accountant works with data all day and does not claim it. Neither does the database engineer who keeps a hospital's records correct, nor the analyst who produces a weekly dashboard of admissions. All three describe what was observed.

The title is most common where the subject matter changes from one quarter to the next. Someone who worked on recidivism last year and on hospital readmission this year gets called a data scientist. An epidemiologist tracing an outbreak may do the same kind of work and never carry the title. The people who claim it work on the methods, designs, and systems that turn observations into evidence. The others use those methods without changing them.

3 Apparatus for Science with Data

Definition 1 (Data science). *Data science* is the field that develops the apparatus for doing science with data. Its practitioners build, adapt, and scrutinize that apparatus, and they use it to support claims about cases, populations, outcomes, and interventions that were not observed.

The ecologist studies bird biodiversity and the astronomer studies a starquake. The data scientist is more likely to study the procedure that finds a stable ecological grouping, or the one that separates a faint stellar signal from instrument noise. A few invent such procedures. Most adapt and combine existing ones for a particular problem, and much of the work is finding out why a procedure that ran fine yesterday has stopped. The roles overlap, and an epidemiologist may build a new causal estimator when the available ones fail.

Data science is often drawn as the intersection of statistics, computer science, and substantive knowledge (Cleveland, 2001; Donoho, 2017). The picture is consistent with Definition 1, though the three parts are rarely equal. A researcher goes to the domain to learn what was measured and how, which question is worth asking, and which assumptions are reasonable. From statistics they get the limits of what the observations can support, and from computer science the means to make the procedure explicit and computable.

A method built for one problem is often used for many, because the method itself is studied. Gosset's small-sample reasoning, described in Section 4, was worked out for brewing and is now

used in agriculture, medicine, and psychology. Other apparatus is only ever used on the problem it was built for, such as a calibration pipeline written for one telescope.

4 Exploring Lineages

The statistical lineage runs from the counting of people to a theory of what a sample can support about a population. The other begins with a word for a recipe, and ends with machines that run recipes at scale. The two ran separately for most of that time.

4.1 The Statistical Lineage

Statistics began as the science of the state, which meant counting people for administration. John Graunt's analysis of the Bills of Mortality showed in 1662 that regularities could emerge from imperfect administrative records (Graunt, 1662). He used the records as he found them, with no say in how they were produced.

The continuous mathematics of variation came from probability, which had been worked out for games of chance and for the errors of astronomical measurement. Gosset worked at Guinness, where experiments were expensive and samples were small, and his 1908 small-sample reasoning became Student's t-test and was taken up well beyond brewing (Student, 1908; Pearson, 1990).

Fisher then set out what the arithmetic was for. In 1922 he wrote that the object of statistical methods is the reduction of data, accomplished by constructing a hypothetical infinite population of which the actual data are regarded as a random sample (Fisher, 1922). The numbers in hand are a draw from a larger population, and the analysis concerns that population. Fisher later made randomization central to experimental design, so that an intervention could be separated from ordinary variation (Fisher, 1935).

By the middle of the twentieth century statisticians had drifted toward mathematical abstraction and away from that idea. In 1962, Tukey argued that data analysis extended beyond mathematical statistics to include examining, transforming, visualizing, and interpreting observations (Tukey, 1962). Wild and Pfannkuch later described statistical thinking in similar terms (Wild and Pfannkuch, 1999). A student can learn a catalog of tests without learning why the data look the way they do.

4.2 The Computational Lineage

Algorithm descends from the name of the ninth-century mathematician al-Khwarizmi, through *algorism*, the medieval term for calculating with Arabic numerals (Knuth, 1997). An algorithm is a recipe, meaning a finite sequence of unambiguous steps that requires no judgement to follow and that terminates with a result.

Turing made the idea exact with his 1936 model of computation. He formalized what it means

to follow such a sequence, and he also established limits on what any such sequence can do (Turing, 1936). The mathematics here is discrete. A step either applies or it does not, and a procedure either halts or it does not.

Knowing that a procedure is exact does not settle where effort should be spent. Turing met that problem at Bletchley Park, where there were more cipher settings than could be tested blindly, and he and his colleagues used partial evidence to decide which ones were worth testing (Good, 1979).

Scale is a further constraint, and decomposition is the usual response. A difficult problem is broken into smaller tasks that can each be specified and repeated, so that a dataset is ingested, checked, transformed, passed through a model, and evaluated. The pieces then have to fit back together into a procedure whose output is still an answer to the original question.

In this lineage a method is judged by what it produces on data that was not used to build it. Donoho describes the common task framework, in which a public training dataset is released, competitors infer a prediction rule from it, and a referee scores submissions automatically (Donoho, 2017). Where researchers had once argued about whether a model was right, they now compared held-out accuracy.

	Statistics	Computer science
The data are	a sample from a population	the input, taken as given
The mathematics is	continuous	discrete
A method is judged by	the adequacy of its model	its score on held-out cases
An explanation is	a mechanism that could have generated the data	a procedure that reproduces the outcome

Table 1: Where the two lineages differ. Neither column defines a field. Both are tendencies data science inherited and has to reconcile.

4.3 The Merger

Breiman stated the difference plainly in 2001. Researchers in one culture model how the data were generated. Those in another treat the mechanism as unknown and judge an algorithm by its predictions (Breiman, 2001). Both lineages are present within statistics, and their commitments conflict.

Answering most real questions takes both. A common task framework requires a scoring rule, which is a statistical object, and data whose origin determines whether the score means anything outside the benchmark. Turing’s decision at Bletchley has the same shape, part procedure and part judgement about what the data support.

Cleveland proposed *data science* in 2001 for a technical field organized around the work of data analysts (Cleveland, 2001). Around 2008, D. J. Patil and Jeff Hammerbacher established *data scientist* as a job title at LinkedIn and Facebook (Donoho, 2017), and a 2012 article in the *Harvard*

Business Review made it famous (Davenport and Patil, 2012). The work was already going on along both lineages, and a name and a labour market came with these events.

5 Features of a Data Scientist

For much of the field's recent history a data scientist's contribution was technical scarcity. Data scientists could query a database, train a neural network, or estimate a causal effect when few collaborators could. Those skills are becoming easier to access as AI tools improve.

The rest of the contribution comes from the two lineages, and it is a way of approaching a problem. A data scientist takes an ill-defined problem, converts it into a formal quantity, devises steps that produce an answer, and says in ordinary language what the answer means. Three kinds of thinking are involved.

Definition 2 (Algorithmic thinking). The decomposition of a task into an ordered sequence of steps that can be repeated, together with analysis of whether those steps produce the intended result and of how their cost grows with the size of the input.

Definition 3 (Statistical thinking). Reasoning about how the data were generated, about which patterns in them are stable, and about how far the observed sample is from the population or setting the claim is about.

Definition 4 (Causal thinking). Reasoning about what a system would do under conditions other than those observed, including actions that were not taken. It requires a formal account of how the system works, a formal statement of the quantity that would answer the question, and an argument that some feature of how the data came about makes that quantity recoverable. Comparing the outcome that followed the action taken with the outcome that would have followed a different action is one part of this rather than the whole of it.

Causal thinking is often introduced too narrowly, as something reserved for decisions. It is needed whenever an inference is meant to be generalizable. Explaining why an outcome occurred is causal even when no one will intervene, and even a descriptive estimate depends on how the sample was selected. The need is most obvious when a decision follows, and it is present even when none does.

Students are taught how to write down the counterfactual, and the surrounding work takes longer. A researcher writes down which variables act on which others, so that the assumptions are visible in a graph or a set of structural equations instead of left tacit (Pearl, 2009). The estimand is the quantity being estimated together with the population it refers to, and the effect of a treatment among those who receive it is a different quantity from its effect in the whole population.

Identification is the next step, and it turns on which features of how the data came about can be exploited. Randomization is one such feature. Others are a rule that assigns treatment at a threshold, a variable that moves treatment without moving the outcome, and a measurement that should show no effect if the design is sound.

6 Data Science amidst Generative AI

Generative AI is used for skill-based work, and writing code is only one example. The output also includes a visualization produced from a verbal request, a fitted model, and a paragraph describing the result. GPT-4 has performed well on some bounded data-analysis tasks (Cheng et al., 2023), and researchers report gains in other forms of knowledge work for at least some workers and tasks (Noy and Zhang, 2023). Performance is uneven. Dell’Acqua and colleagues call the pattern a jagged frontier, where the same system is useful inside one region of tasks and wrong just outside it, with nothing in the output to mark the difference (Dell’Acqua et al., 2023).

A skill, here, is the ability to execute a specified procedure. Writing a merge, tuning a model, and drawing a figure to a stated specification are all skills, and in each case the specification arrives from somewhere else. Input of that kind suits these systems, and their performance on it improves every year. An account of the occupation built only on skills is therefore vulnerable.

The three kinds of thinking in Section 5 work the other way round. A person uses algorithmic thinking to arrive at what the procedure should be, statistical thinking to establish whether its output can be believed for a case that was not observed, and causal thinking to establish what would have to be true for the answer to hold outside the data at hand. In each case the specification is an output of that work, not an input to it, and reading the result gives no sign of whether the work was done.

In Turing’s imitation game a person tried to distinguish a machine’s written responses from another person’s (Turing, 1950). Fluency is a poor test of an analysis, because a generated analysis and a correct one look the same on the page. Someone has to establish that the code implements the intended procedure and that the study design supports the conclusion.

None of this is an argument against learning skills. A person learns what a confounder does by fitting the model with it and without it, so the skill is how the thinking gets built. Stopping once the skill is acquired no longer works.

Executing a procedure will be worth less, and choosing which procedure answers the question will be worth more. A curriculum organized around software tutorials will age quickly. Students need to learn how to break a problem apart, interrogate where the data came from, and put a decision’s question into a form that can be answered.

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